Exp No: 4

Date: 30/7/26 setup and configure a LAN (local area network) using a switch and Ethernet cables in your lab.

Aim:

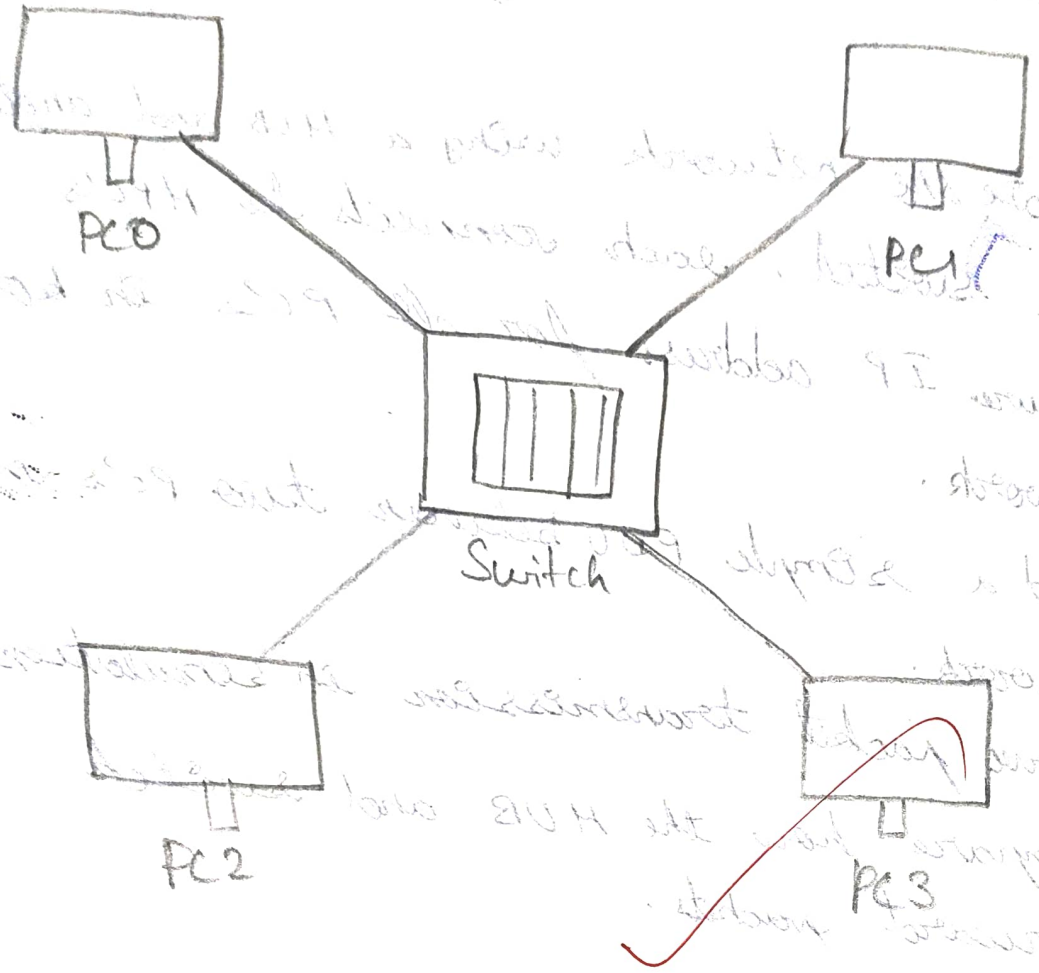
To setup and configure a LAN (local area network) using a switch and Ethernet cables in your lab.

Steps to setup AN LAN:-

step 1: plan and design an appropriate network topology taking into account network requirements and equipment location

step 2: Take 4 computers, a switch with 8, 16 or 24 ports which is sufficient for network of these size and 4 Ethernet cables.

step 3: connect the computers to network switch via an ethernet cable, which is as simple as plugging one end of the Ethernet cable into the computer and the other end into the network switch.


PC0
PC1
Switch
PC2
PC3

step 4: Assign IP address to your PC's

(1.) log on the client computer as Administrator or as owner.

(2.) click Networks and Internet connections.

(3.) Right click local area connection / Ethernet -> go to properties -> select Internet protocol (TCP/IPV4) -> click on properties -> select use the following ip address option and assign ipaddress.

step 5: configure a network switch:

(1.) connect the computer to the switch:
To access the switch's web

(2.) log in to the web interface

(3.) configure basic settings

(4.) Assign IP address as 10.10.1.5, subnet mask 255.0.0.0.

step 6: check the connectivity between switch and other machine by using ping command in the command prompt of the same LAN.

step 7:- Select a folder -> go to properties -> click sharing tab -> share it with everyone on the same LAN.

step 8:

Try to access the shared folder from other computers of the network.

Result:

A local area network (LAN) was successfully setup using one Network Switch, four computers and the Ethernet cables.